

Security & privacy in FW 5

Xovis sensors and data protection

Data privacy and data security are global concerns. Most countries have laws in place to protect personal data from misuse and destruction. The details and impact are country specific, vary greatly and must be assessed by each company individually.

The aim of regulations such as the General Data Protection Regulation in Europe is to reinforce data protection rights of individuals, facilitate the free flow of personal data in the digital single market and reduce the administrative burden of managing personal data.

Anything that can personally identify someone, be it their name, address, financial details, or posts across social media, is covered by the regulation. As a result, more data than ever before will fall under protection legislation.

While Xovis sensors provide data for person counting and tracking, specific technical functions are offered to provide data securely and in a de-personalized manner.

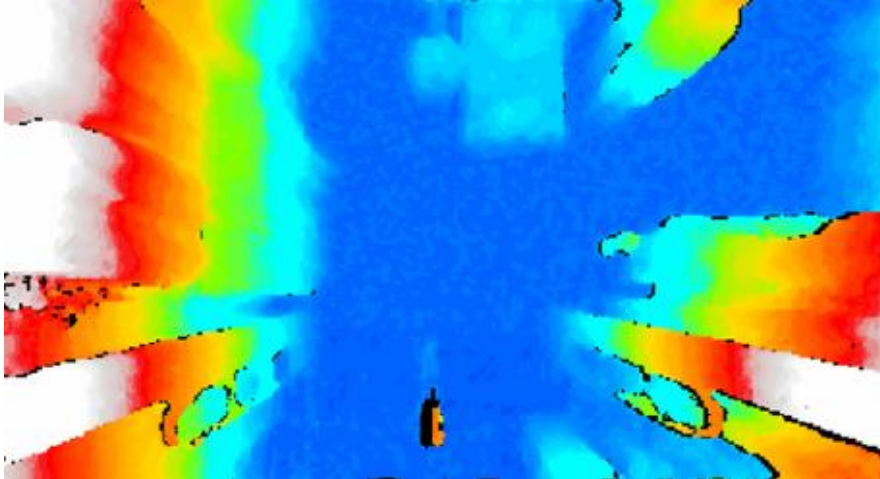
Providing Security/Privacy

General settings

Xovis sensors are network-based devices. Many security features are provided to set up a sensor for compliance with local data privacy laws and IT security requirements. The person in charge of installation is obligated to follow all state-of-the-art cyber security implementations to place Xovis sensors in a protected and trusted network environment.

Image processing

The functionality of Xovis 3D stereo vision sensors is based on two optical CMOS image sensors arranged at a certain distance to each other. The images are compiled to a stereo image in real time on the device. The processed images are not stored. Neither will these images leave the sensor at any time during operation. Only metadata, such as count metrics, timestamp is pushed from the sensor.



Password protection

Access to the sensor is password protected. Setting a strong individual password is the basic security measure for all sensors.

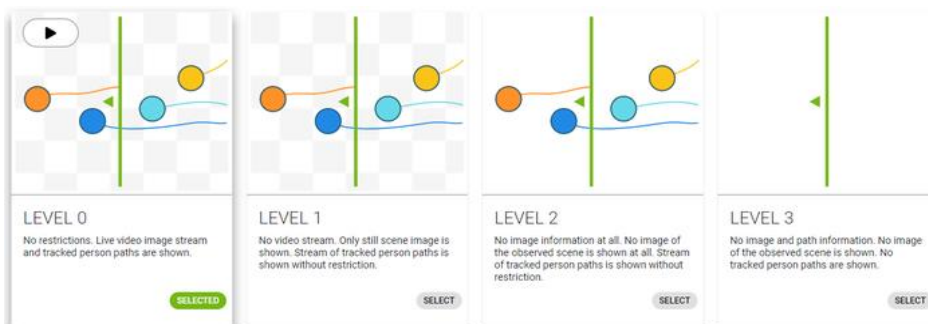
Change admin password 

Enable viewer account ☐

Privacy level settings

Xovis sensors provide several types of data output streams. With the privacy settings available on all sensors, visual images can be deactivated completely. The privacy mode level can be increased at any time.

Attention: The Sensor Master Key (SMK) is needed to lower the privacy level. The SMK is a unique key created for each sensor.



Level	Description
0	No restrictions. A live video of the scene as well as person tracking is shown without any restrictions in the WebUI.
1	Only a still image of the scene is shown, no video stream. This still image can be refreshed on request. Tracking is shown without restrictions.
2	No image information is shown at all. The scene background is just displayed as a gray area. Tracking is shown without restrictions.
3	No image information and no tracking information is shown. Only the counting values are updated.

Ports & SSL certificates

Xovis sensors provide the functionality to control the network ports such as HTTP(s).

HTTP *
☒

HTTPS *
☒

Manage ports

Every port the sensor uses can be changed and most of the services can be disabled. HTTP can be disabled permanently to prevent any non-encrypted access to the sensor API and WebUI.

Manage server identity

Custom SSL keys and certificates can be uploaded and applied to the sensor. This overwrites the Xovis default key/certificate or any previous server identity.

Manage trusted authorities

Users can manage their own trusted authorities. Certificates can be uploaded and will automatically be applied. Any uploaded certificate can be removed at any time. The Xovis default certificate authority (CA) can be enabled and disabled.

Xovis recommends:

- Put a strong password according a company's policy and commonly best practices (which might change from time-to-time and are to find via Google)
- Place the internal network and sub-nets behind a firewall

Hashing algorithm security for WiFi/BLE monitoring

Dedicated Xovis sensors can monitor WiFi and Bluetooth signals from devices in the vicinity. An URL can be specified to identify a server which gathers the IDs of all detected wireless devices in a defined time interval. Each item contains the device ID (MAC), the signal strength in dB, and the corresponding device type (Bluetooth, Bluetooth low energy, or WiFi). In case of the detection of a Bluetooth device, its UUID is transmitted additionally.

The sensor settings provide the possibility to

- turn the monitoring feature on and off
- set up a Allowlist
- set up a Denylist

With an Allowlist in use, only listed devices get monitored by the sensor. Devices listed in a Denylist are not being monitored by the sensor.

The legal operation of this feature depends on specific country regulations.

Example: In Germany it is not allowed to place WiFi/Bluetooth monitoring systems in public areas. It is the user's responsibility to assure legal compliance of all Xovis devices.

API

The API for sensor firmware >5.0 is no more public except of some calls, which help to find installed sensors in a network. All other API need authentication.

Additional built-in security measures

- Rate limiter for authentications
- Remote connections only allowed over encrypted TLS connections
- Reworked security concept to prevent signal level serial console access
- Secure boot chain to ensure untampered TLS certificates and other cryptographic keys
- Multisensor connections can be via https
- Diagnostic file (.xdg) encrypted
- Sensor backup file (.xbak) encrypted

GDPR

Xovis sensors allow to be operated in a GDPR-compliant system.

Important: The overall system needs to be compliant, not only the sensor.

The system provider is responsible to deal with the needed **technical and organizational measures (TOM)**, which means among others:

- Controlling sensor access
- Controlling access to SMK
- Defining data processing and storage

More details can be found in our declaration of product policy – as attached **declaration-of-product-data-privacy.pdf**